

Venom RAT detection with Wazuh

(Remote Access Trojan)

Venom RAT (Remote Access Trojan) is a type of malicious software that allows an attacker to take remote control of a victim's computer without their knowledge or consent. Once installed on the victim's computer, the attacker can perform a variety of actions such as stealing sensitive information, monitoring user activity, and using the infected computer as part of a botnet to launch further attacks.

Venom RAT can be distributed through a variety of methods, including phishing emails, malicious websites, and software vulnerabilities. It is important to have a good antivirus program and to keep your operating system and software up-to-date to minimize the risk of becoming infected with Venom RAT or other malware. It is also important to be cautious when clicking on links or downloading attachments from unknown sources, and to avoid entering sensitive information on unsecured websites.

Venom RAT behavior:

- ➔ It drops a copy of itself in Path with svchost.exe to masquerade as a legitimate file.
- ➔ Venom RAT creates a registry key for the created svchost.exe file to ensure it runs anytime the victim endpoint restarted.
- ➔ Venom RAT injects itself into multiple .dll files.
- ➔ Venom RAT creates .tmp and .tmp.bat files.
- ➔ Venom RAT executes the created .tmp.bat file.
- ➔ Venom RAT uses cmd.exe to delete the created .tmp.bat file.

Windows Endpoint

- ➔ This configuration is same as of previous Use Case of Black cat ransomware. You can refer to that use case.

Wazuh server

- ➔ Add the rules below to the `/var/ossec/etc/rules/local_rules.xml` file on the Wazuh server:

```
GNU nano 6.2 /var/ossec/etc/rules/local_rules.xml
<group name="venom_rat,syscheck">

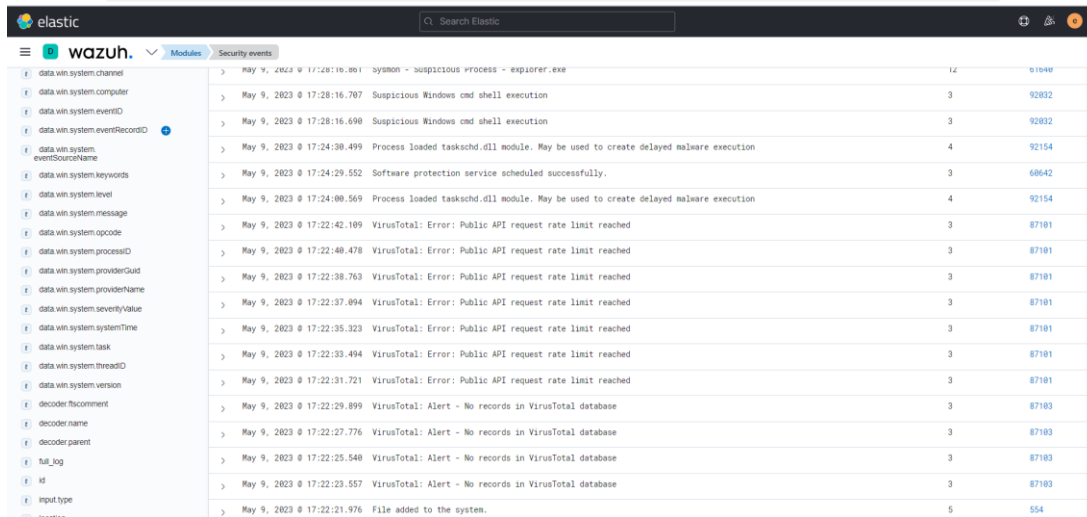
  <!-- Rogue svchost.exe creation -->
  <rule id="100950" level="12">
    <if_sid>61613</if_sid>
    <field name="win.eventdata.image" type="pcre2">\\.exe</field>
    <field name="win.eventdata.targetFilename" type="pcre2">(?!)[c-z]:\\\\Users\\\\.+\\\\AppData\\\\Roaming\\\\svchost\\.exe</field>
    <description>Potential Venom RAT activity detected: svchost.exe created at $(win.eventdata.targetFilename) by $(win.eventdata.image).</description>
    <mitre>
      <id>T1036</id>
    </mitre>
  </rule>

  <!-- Registry key creation for persistence -->
  <rule id="100951" level="12">
    <if_sid>92300</if_sid>
    <field name="win.eventdata.image" type="pcre2">(?!)[c-z]:(\\\\Users\\\\.+\\\\)\\</field>
    <field name="win.eventdata.details" type="pcre2">(?!)[c-z]:\\\\Users\\\\.+\\\\AppData\\\\Roaming\\\\svchost\\.exe</field>
    <description>Potential Venom RAT activity detected: $(win.eventdata.details) added itself to the Registry as a startup program to establish persistence.
    <mitre>
      <id>T1547.001</id>
    </mitre>
  </rule>

  <!-- Suspicious .tmp.bat file creation -->
  <rule id="100952" level="12">
    <if_sid>92204</if_sid>
    <field name="win.eventdata.image" type="pcre2">\\.exe</field>
    <field name="win.eventdata.targetFilename" type="pcre2">(?!)[c-z]:\\\\Users\\\\.+\\\\AppData\\\\Local\\\\Temp\\\\.+\\.tmp\\.bat</field>
    <description>Potential Venom RAT activity detected: Suspicious .tmp.bat file $(win.eventdata.targetFilename) added to Temp folder by $(win.eventdata.image).
    <mitre>
      <id>T1059</id>
    </mitre>
  </rule>

  <!-- Suspicious .tmp.bat file run in command line-->
  <rule id="100953" level="15">
    <if_sid>61603</if_sid>
    <field name="win.eventdata.parentImage" type="pcre2">(?!)[c-z]:\\\\cmd\\.exe</field>
    <field name="win.eventdata.parentCommandLine" type="pcre2">(?!)[c-z]:\\\\s\\\\C\\\\s\\.\\.\\.tmp\\.bat</field>
    <description>Venom RAT activity detected: Suspicious .tmp.bat file executed by cmd.</description>
    <mitre>
      <id>T1087</id>
      <id>T1059.003</id>
    </mitre>
  </rule>
</group>
```

- ➔ Restart the Wazuh manager to apply the configuration changes:
systemctl restart wazuh-manager



data.win.system.channel	>	May 9, 2023 @ 17:28:10.801	symon - suspicious process - explorer.exe	12	61640
data.win.system.computer	>	May 9, 2023 @ 17:28:16.707	Suspicious Windows cmd shell execution	3	92832
data.win.system.eventID	>	May 9, 2023 @ 17:28:16.690	Suspicious Windows cmd shell execution	3	92832
data.win.system.eventRecordID	>	May 9, 2023 @ 17:24:30.499	Process loaded taskshd.dll module. May be used to create delayed malware execution	4	92154
data.win.system.eventSourceName	>	May 9, 2023 @ 17:24:29.552	Software protection service scheduled successfully.	3	68642
data.win.system.keywords	>	May 9, 2023 @ 17:24:00.569	Process loaded taskshd.dll module. May be used to create delayed malware execution	4	92154
data.win.system.level	>	May 9, 2023 @ 17:22:42.109	VirusTotal: Error: Public API request rate limit reached	3	87101
data.win.system.message	>	May 9, 2023 @ 17:22:40.478	VirusTotal: Error: Public API request rate limit reached	3	87101
data.win.system.opcode	>	May 9, 2023 @ 17:22:38.763	VirusTotal: Error: Public API request rate limit reached	3	87101
data.win.system.processID	>	May 9, 2023 @ 17:22:37.094	VirusTotal: Error: Public API request rate limit reached	3	87101
data.win.system.providerGuid	>	May 9, 2023 @ 17:22:35.323	VirusTotal: Error: Public API request rate limit reached	3	87101
data.win.system.providerName	>	May 9, 2023 @ 17:22:33.494	VirusTotal: Error: Public API request rate limit reached	3	87101
data.win.system.severityValue	>	May 9, 2023 @ 17:22:31.721	VirusTotal: Error: Public API request rate limit reached	3	87101
data.win.system.systemTime	>	May 9, 2023 @ 17:22:29.899	VirusTotal: Alert - No records in VirusTotal database	3	87103
data.win.system.task	>	May 9, 2023 @ 17:22:27.776	VirusTotal: Alert - No records in VirusTotal database	3	87103
data.win.system.threadID	>	May 9, 2023 @ 17:22:25.540	VirusTotal: Alert - No records in VirusTotal database	3	87103
data.win.system.version	>	May 9, 2023 @ 17:22:23.557	VirusTotal: Alert - No records in VirusTotal database	3	87103
decoder.flcomment	>	May 9, 2023 @ 17:22:21.976	File added to the system.	5	554

Thank you happy hunting 😊